

Task 1: Basic Network Scanning with Nmap

Objective

Perform a network scan to identify open ports and services running on a local machine using Nmap.

Tools Used

- Nmap 7.80 — Network scanning tool
- OS: Windows (localhost scan)

Scan Command Used

```
bash
nmap -sV -oN nmap_scan_results.txt 127.0.0.1
```

Flag Explanation

Flag	Meaning
-sV	Detect service versions on open ports
-oN	Save output to a file in normal format
127.0.0.1	Target IP — localhost (this machine)

Scan Summary

- Target: 127.0.0.1 (localhost / kubernetes.docker.internal)
- Host Status: Up (latency: 0.000077s)
- Total Ports Scanned: 1000 (default)
- Closed Ports: 996
- Open Ports Found: 4
- Scan Duration: 18.77 seconds
- Date: Wed May 27, 2026

Open Ports & Services Found

Port	State	Service	Version	Risk Level
135/tcp	Open	Microsoft RPC	Windows MSRPC	Medium
445/tcp	Open	SMB	Microsoft-DS	High
1434/tcp	Open	SQL Server Browser	tcpwrapped	Medium
3306/tcp	Open	MySQL	MySQL 8.0.44	High

Detailed Port Analysis

Port 135 — Microsoft RPC (Remote Procedure Call)

- **What it is:** A core Windows service that allows programs to communicate over a network.
- **Why it's open:** Required by Windows for services like DCOM, WMI, and Active Directory.
- **Security Concern:** Historically targeted by worms (e.g., Blaster worm in 2003). Should be blocked on public-facing systems. On localhost it is normal.
- **Recommendation:** Block port 135 on the firewall for external/public interfaces.

Port 445 — SMB (Server Message Block)

- **What it is:** Used for Windows file sharing, printer sharing, and network communication.
- **Why it's open:** Windows enables SMB by default for local network file sharing.
- **Security Concern:** This is one of the most commonly exploited ports. The **EternalBlue exploit** (used in WannaCry ransomware, 2017) targeted this port. Running on localhost is normal, but it must never be exposed to the internet.
- **Recommendation:** Disable SMBv1, ensure SMB is blocked at the perimeter firewall, keep Windows updated.

Port 1434 — Microsoft SQL Server Browser

- **What it is:** SQL Server Browser service — helps clients discover SQL Server instances on the network.
- **Why it's open:** Installed alongside Microsoft SQL Server or certain developer tools.
- **Security Concern:** Marked as `tcpwrapped`, meaning the service accepted a connection but did not respond with identifiable data — this is a security measure. Still, unnecessary

exposure can lead to reconnaissance attacks.

- **Recommendation:** Disable SQL Server Browser if not needed. Restrict access to trusted IPs only.

Port 3306 — MySQL Database

- **What it is:** MySQL is a popular open-source relational database. Port 3306 is its default port.
- **Why it's open:** A local MySQL 8.0.44 instance is running — likely installed with XAMPP, MySQL Workbench, or a development environment.
- **Security Concern:** High risk if exposed externally. The scan fingerprint confirms **MySQL 8.0.44** with `caching_sha2_password` authentication (a modern, more secure auth plugin). However, databases should never be publicly accessible.
- **Recommendation:** Bind MySQL to `127.0.0.1` only (already default). Never expose port 3306 to the internet. Use strong passwords and limit user privileges.

Key Findings Summary

1. This machine is a **Windows development machine** running Docker (`kubernetes.docker.internal`), SQL Server tools, and MySQL.
2. All open ports are **expected for a developer environment** and are only on localhost (`127.0.0.1`), so external risk is low.
3. **Port 445 (SMB)** is the most sensitive port — it must remain blocked on any external firewall.
4. **Port 3306 (MySQL)** is confirmed running MySQL 8.0.44 — should never be exposed beyond localhost.
5. No unexpected or suspicious services were found.

Security Recommendations

Port	Action
135	Block on external firewall; normal on localhost
445	Ensure SMBv1 is disabled; block externally
1434	Disable SQL Server Browser if not in use
3306	Keep MySQL bound to 127.0.0.1 only; use strong passwords

Files in This Repository

File	Description
<code>nmap_scan_results.txt</code>	Raw Nmap scan output
<code>README.md</code>	This file — analysis and explanation of results
<code>screenshots/</code>	Screenshots of Nmap running and output

Conclusion

This task demonstrated how to use Nmap to perform a basic service version scan (`-sV`) on a local machine. Four open ports were identified — all consistent with a Windows developer environment. Understanding what each port does and its associated security risks is a fundamental skill in network security and penetration testing.

Scan performed as part of Cybersecurity Internship — Task 1: Basic Network Scanning with Nmap